The Arcyon International Institute of Medical Science & Technology Council

Indian Social Institute, Bengaluru, 560046

Domain: IOT and Embedded System

Project Report on

"Iot-Based Smart Walking Cane For Obstacle Detection And Safety Enhancement "

Submitted by:

Cohort 5

Under the Guidance of:

Dr. Padma Giri
Principal & Academic Director
AIIMS Technology Council

IIMSTC, BENGALURU

Indian Social Institute, 24, Benson Rd, Byadarahalli, Benson Town, Bengaluru, Karnataka 560046

03/02/2026 – 20/05/2026

ABSTRACT

Navigation and mobility remain significant challenges for visually impaired individuals, as traditional assistive tools such as white canes provide limited functionality and detect obstacles only after physical contact. This delayed detection increases the risk of collisions, falls, and injuries caused by obstacles, uneven surfaces, and environmental hazards. To overcome these limitations, this project proposes an IoT-based Smart Walking Cane for Obstacle Detection and Safety Enhancement, designed to provide early hazard detection and real-time assistance. The proposed system integrates multiple sensors, including an ultrasonic sensor for obstacle detection, an infrared (IR) sensor for pit and edge detection, a light-dependent resistor (LDR) for low-light detection, and a water sensor for identifying wet or slippery surfaces. These sensors continuously monitor the surroundings and transmit data to an ESP32 microcontroller, which processes the inputs and determines appropriate actions. When a hazard is detected, the system provides immediate feedback to the user through a buzzer, enabling safer navigation. In addition to real-time alerts, the system incorporates a Global Positioning System (GPS) module to track the user's location. The collected data is transmitted to a cloud platform such as Firebase using Wi-Fi connectivity. This enables remote monitoring of the user's location and system status. An emergency (SOS) feature is also included to enhance safety during critical situations by sending alerts to caregivers. The system is designed to be cost-effective, portable, and energy-efficient, making it suitable for real-world applications. Experimental results demonstrate reliable obstacle detection, effective hazard identification, and efficient cloud communication. Overall, the proposed smart cane improves the safety, independence, and quality of life of visually impaired individuals while providing a scalable solution for future enhancements.

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Visual impairment is a major global concern that significantly affects an individual's ability to navigate safely and independently. According to global health studies, millions of people rely on assistive devices such as white canes for mobility. While the traditional white cane is a simple and affordable tool, it has several limitations in detecting obstacles and environmental hazards.

A conventional cane primarily depends on physical contact to identify obstacles, which means that users become aware of hazards only after touching them. This delayed detection can lead to accidents such as collisions, falls, or injuries, especially in complex environments like crowded streets, staircases, or uneven surfaces.

With advancements in embedded systems, sensors, and Internet of Things (IoT) technologies, it is now possible to enhance traditional assistive devices by integrating smart features. Modern smart canes use sensors to detect obstacles, microcontrollers to process data, and communication modules to provide real-time feedback and monitoring.

The integration of IoT platforms allows these devices to transmit data to cloud systems, enabling remote monitoring and improved safety. This project leverages these technologies to develop a smart walking cane that provides early hazard detection, real-time alerts, and location tracking.

1.2 PROBLEM STATEMENT

Despite the availability of traditional assistive devices, visually impaired individuals still face significant challenges in navigating safely. The key problems associated with existing systems are:

- Traditional canes detect obstacles only after physical contact
- Lack of real-time feedback mechanisms
- Inability to detect hazards such as water, pits, or low-light conditions
- No support for emergency communication or location tracking
- Limited integration with modern IoT technologies

These limitations increase the risk of accidents and reduce the independence of visually impaired individuals. Therefore, there is a need for an intelligent, cost-effective, and reliable system that can detect obstacles in advance, provide real-time alerts, and enable remote monitoring.

1.3 OBJECTIVES

The main objectives of this project are:

- To design and develop a smart walking cane using IoT technology
- To detect obstacles using ultrasonic sensors before physical contact
- To identify environmental hazards such as water and pits using appropriate sensors
- To provide real-time alerts through buzzer or vibration mechanisms
- To integrate GPS for real-time location tracking
- To enable cloud-based monitoring using IoT platforms
- To improve the safety, mobility, and independence of visually impaired individuals

1.4 SCOPE OF THE PROJECT

The scope of this project includes the design, development, and implementation of a smart assistive cane capable of detecting obstacles and hazards in real-time. The system is intended for both indoor and outdoor environments, providing users with enhanced navigation support.

The project focuses on:

- Real-time obstacle detection using sensors
- Hazard identification including water and uneven surfaces
- Alert generation through sound or vibration
- Location tracking using GPS
- Data transmission to cloud platforms for monitoring

However, the project does not include advanced features such as computer vision or AI-based object recognition, which can be considered for future enhancements.

1.5 MOTIVATION

The primary motivation behind this project is to improve the quality of life for visually impaired individuals by providing a safer and more reliable navigation system. Observing the limitations of traditional canes and the risks faced by users highlights the need for technological intervention.

This project aims to bridge the gap between basic assistive devices and modern smart systems by incorporating affordable and efficient technologies. The use of IoT, sensors, and embedded systems makes it possible to develop a solution that is both practical and scalable.

1.6 APPLICATIONS

The proposed system has several practical applications, including:

- Assistive device for visually impaired individuals
- Navigation support in indoor and outdoor environments
- Smart healthcare and rehabilitation systems
- Integration into wearable assistive technologies

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW

Recent studies in assistive technology have demonstrated significant advancements in the development of smart walking canes through the integration of Internet of Things (IoT) and sensor-based systems. These innovations aim to address the limitations of traditional mobility aids and provide enhanced safety, navigation, and independence for visually impaired individuals. Researchers have focused on combining multiple technologies such as ultrasonic sensors, water sensors, GPS modules, and communication systems to create intelligent and responsive assistive devices.

Sensor integration plays a crucial role in enhancing the functionality of smart canes. Ultrasonic sensors are widely used for detecting obstacles at various distances, including those above ground level, which traditional canes fail to identify. Some studies also incorporate infrared sensors and advanced detection techniques to improve accuracy and reliability. Additionally, water sensors are used to detect wet or slippery surfaces, helping to prevent accidents caused by falls. Certain designs further include features for detecting stairs or uneven terrain, providing comprehensive environmental awareness.

Navigation and alert systems have also been significantly improved in modern smart cane designs. The inclusion of Global Positioning System (GPS) modules allows real-time location tracking, which is particularly useful during emergencies. Many research works highlight the use of GSM or Bluetooth modules for sending alerts and notifications to predefined contacts. Feedback mechanisms such as buzzers and vibration motors are commonly used to convey warnings to the user in an intuitive and non-visual manner. Different alert patterns can indicate the type or distance of obstacles, making the system more user-friendly.

Overall, the literature indicates that IoT-based smart walking canes offer a promising solution to enhance the safety and independence of visually impaired individuals. By integrating advanced sensors, real-time communication, and intelligent feedback systems, these devices overcome the limitations of traditional canes. Continuous research and development in this field are expected to further improve accuracy, affordability, and usability, making smart assistive technologies more accessible to a wider population.

2.2 RESEARCH PAPERS

1. The research paper titled "Design and Development of a Smart Cane Using Ultrasonic Sensors and GPS for Visually Impaired People" was published in the African Journal of Advanced Pure and Applied Sciences (AJAPAS), Volume 4, Issue 4, 2025, pages 599–605, by authors Amier A. Frag, Omran A. H. Mohamed, Massoud A. Omar Imaezeg, and Yaser S. A. Shaheen. The paper focuses on developing a smart assistive cane for visually impaired individuals using ultrasonic sensors, GPS technology, and a microcontroller-based system to improve safe and independent mobility. The proposed smart cane detects obstacles within a range of up to 3 meters in front and side directions and alerts the user through vibration and sound feedback mechanisms. The system integrates components such as Arduino Uno/Nano, HC-SR04 ultrasonic sensors, IR sensors, GPS module, Bluetooth module, vibration motor, buzzer, and rechargeable Li-ion battery for efficient operation. The working mechanism involves ultrasonic sensors continuously transmitting sound waves and measuring the reflected echo time to calculate the obstacle distance using the formula: \text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2} The microcontroller analyzes the detected distance and generates alerts depending on the danger level, while the GPS module provides real-time location tracking and emergency notifications during falls or emergencies. The prototype was experimentally tested with 8 participants, including visually impaired users and blindfolded individuals, across multiple environments such as indoor areas, outdoor sidewalks, stairways, uneven surfaces, and crowded locations. The results demonstrated high performance with 95.4% front obstacle detection accuracy, 91.2% side obstacle detection accuracy, 87.6% stair detection accuracy, and an average response time of 0.4 seconds. User feedback indicated that vibration alerts were highly effective and improved confidence during navigation. The study concluded that the proposed smart cane is a low-cost, reliable, and efficient assistive technology solution that significantly enhances the safety, independence, and mobility experience of visually impaired individuals, although some limitations were observed in rainy environments and reflective surfaces.
2. The paper titled "Smart Assistive System for Visually Impaired People Obstruction Avoidance Through Object Detection and Classification" was published in IEEE Access, Volume 10, in 2022 by Usman Masud, Tareq Saeed, Hunida M. Malaikah, Fezan Ul Islam, and Ghulam Abbas. The research focuses on developing a smart assistive navigation system for visually impaired individuals using advanced object detection and obstacle avoidance technologies. The proposed system integrates Raspberry Pi 4B, Arduino Uno, ultrasonic

sensors, Raspberry Pi camera, servo motor, buzzer, battery, TensorFlow Object Detection API, and Viola-Jones algorithm to identify obstacles and classify surrounding objects in real time. The ultrasonic sensor mounted on a servo motor continuously scans left, right, and front directions to detect obstacles and measure their distance, while different beeping sounds are used to indicate obstacle positions. Simultaneously, the camera captures scene images which are preprocessed using Gaussian Blur and segmentation techniques to remove noise and improve image quality before classification. The Viola-Jones algorithm is used for face and object detection through Haar feature selection, integral image generation, AdaBoost training, and cascaded classifiers, while TensorFlow Object Detection identifies multiple objects such as suitcases, keyboards, backpacks, bottles, cell phones, chairs, and beds with high efficiency. The system uses the COCO 2017 dataset containing around 123,000 images for training and achieved an average detection accuracy of approximately 91%, with some objects such as suitcases and backpacks detected with 97% accuracy. The device provides audio and haptic feedback to guide visually impaired users safely and works without continuous internet connectivity since all data processing occurs on the Raspberry Pi itself. The researchers also highlighted advantages such as low cost, portability, rechargeable 24-hour battery backup, real-time obstacle detection, and easy wearable design. However, the paper also mentions limitations including reduced accuracy due to camera shaking, lighting conditions, motion blur, and difficulties in detecting objects that are out of frame or colorless. The study concludes that the proposed smart assistive system significantly improves mobility, safety, and independence for visually impaired people and can be further enhanced in future by integrating GSM modules and text-to-speech systems for voice navigation and emergency communication.

3. The research paper titled "Smart Blind Stick using Mobile Application and Bluetooth Module" was published in the International Journal of Industry and Sustainable Development (IJISD), Volume 6, Issue 1, January 2025, by Shimaa Mahdy, Ibrahim Yousri, Mohammed Hamady, Mohamed Ashraf, Ayman Morad, and Yasser Elshrief from the Department of Electrical Engineering, Egyptian Academy for Engineering and Advanced Technology, Cairo, Egypt. The paper presents an advanced smart blind stick designed to assist visually impaired individuals in safely navigating their surroundings by detecting obstacles such as walls, sidewalks, stairs up, stairs down, holes, water puddles, motion, and dark areas. The proposed system uses Arduino Mega 2560 as the main controller and integrates four HC-SR04 ultrasonic sensors, PIR sensor, IR sensor, rain sensor, LDR sensor, Bluetooth module HC-05, buzzer, vibration motor, flashlight, mobile application, and rechargeable Lithium-Ion battery. The ultrasonic sensors calculate obstacle distance using

reflected sound waves and are strategically mounted at different heights and angles on the stick to improve detection accuracy, while the IR sensor activates the ultrasonic sensors only when the stick is positioned correctly to increase measurement reliability. The PIR sensor detects nearby motion or people, the rain sensor identifies water puddles, and the LDR sensor detects dark areas and automatically activates a flashlight to alert surrounding pedestrians. The Bluetooth-connected mobile application plays a major role by providing voice guidance messages for each obstacle type, sending emergency SMS messages containing the user’s GPS location to friends or family, displaying battery status, customizing feedback settings, and helping locate the stick through buzzer activation. The obstacle detection logic is based on distance threshold calculations between multiple ultrasonic sensors for distinguishing walls, sidewalks, upstairs, downstairs, and holes. The proposed system achieved an average performance accuracy of 98%, where wall, sidewalk, downstairs/hole, light, and water detection achieved 100% success, while upstairs and motion detection achieved 93% success during multiple testing trials. The rechargeable battery provides approximately 2.5 to 3.5 hours of operation depending on usage. The paper highlights advantages such as low cost, portability, lightweight structure, high accuracy, low power consumption, easy maintenance, emergency support, and improved safety for visually impaired users. However, the study mentions one limitation that the system cannot completely differentiate between downstairs and holes, which the authors plan to improve in future work using additional ultrasonic sensors and enhanced threshold calculations.

2.2 EXISTING TECHNOLOGIES

Existing smart cane technologies utilize ultrasonic and infrared sensors, AI-based systems, and IoT-connected devices to detect obstacles and assist navigation. Modern solutions combine multiple sensors, GPS, and mobile connectivity to enhance safety, real-time tracking, and user interaction.

2.3 LIMITATIONS OF EXISTING SYSTEMS

Existing systems often suffer from high cost, limited accuracy in detection, and poor battery performance, reducing their practicality for daily use. Additionally, many solutions lack proper integration of features, leading to inefficient performance and reduced reliability for users.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 INTRODUCTION

This chapter explains the overall architecture and design of the proposed IoT-based smart walking cane. The system is developed to assist visually impaired individuals by detecting obstacles and environmental hazards, providing real-time alerts, and enabling remote monitoring through cloud integration. The architecture is designed in a modular way so that each component performs a specific function efficiently.

3.2 SYSTEM OVERVIEW

The proposed system consists of four main parts:

- Sensing Unit – collects data from the environment
- Processing Unit – processes the sensor data
- Alert Unit – notifies the user about hazards
- Communication Unit – sends data to the cloud

All sensors are connected to the ESP32 microcontroller, which acts as the central control unit. Based on the input from sensors, the system takes appropriate action such as activating the buzzer or sending data to the cloud.

3.3 BLOCK DIAGRAM

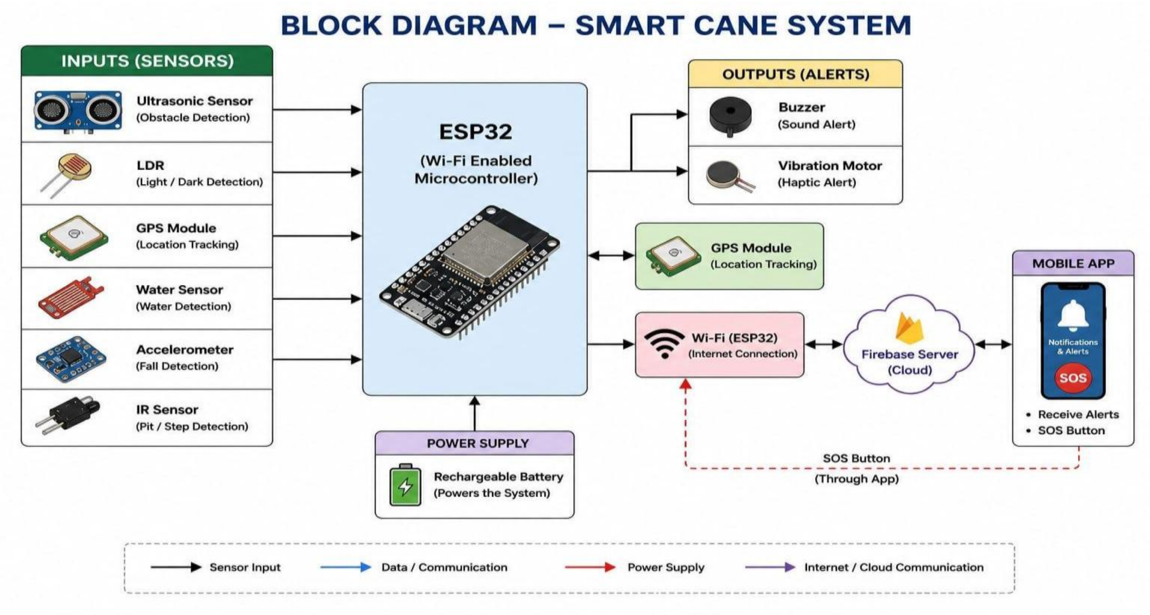

Fig. Block Diagram

3.4 DETAILED DESCRIPTION OF SYSTEM COMPONENTS

3.4.1 SENSING UNIT

The sensing unit is responsible for collecting data from the surrounding environment. Multiple sensors are used to improve accuracy and reliability:

- **Ultrasonic Sensor:**
  This sensor detects obstacles in front of the user by measuring distance. It sends ultrasonic waves and calculates the distance based on the echo received. It helps in detecting objects before physical contact.

Fig. Ultrasonic Sensor

- **Infrared (IR) Sensor:**
  The IR sensor is used to detect pits, stairs, or edges. It works based on infrared reflection from the ground surface. If there is no reflection, it indicates a gap or pit.

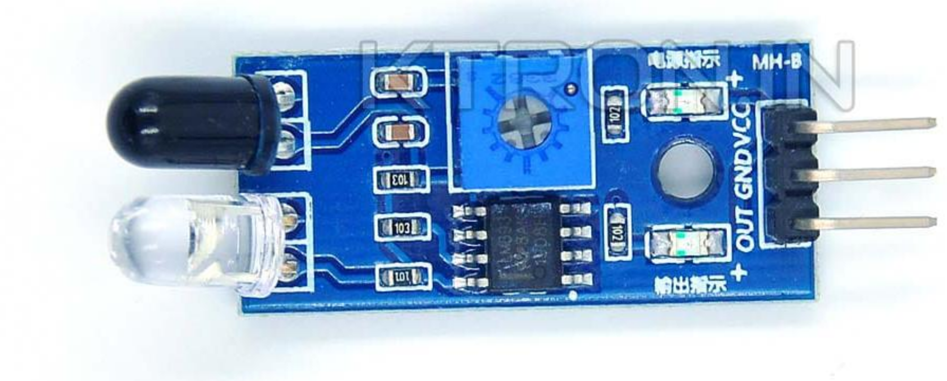

Fig. Infrared (IR) Sensor

- **LDR (Light Dependent Resistor):**
  The LDR sensor detects light intensity. It helps identify low-light or dark environments, improving awareness in poorly lit areas.

Fig. LDR

- **Water Sensor:**
  This sensor detects the presence of water or wet surfaces. It helps prevent slipping accidents by alerting the user.

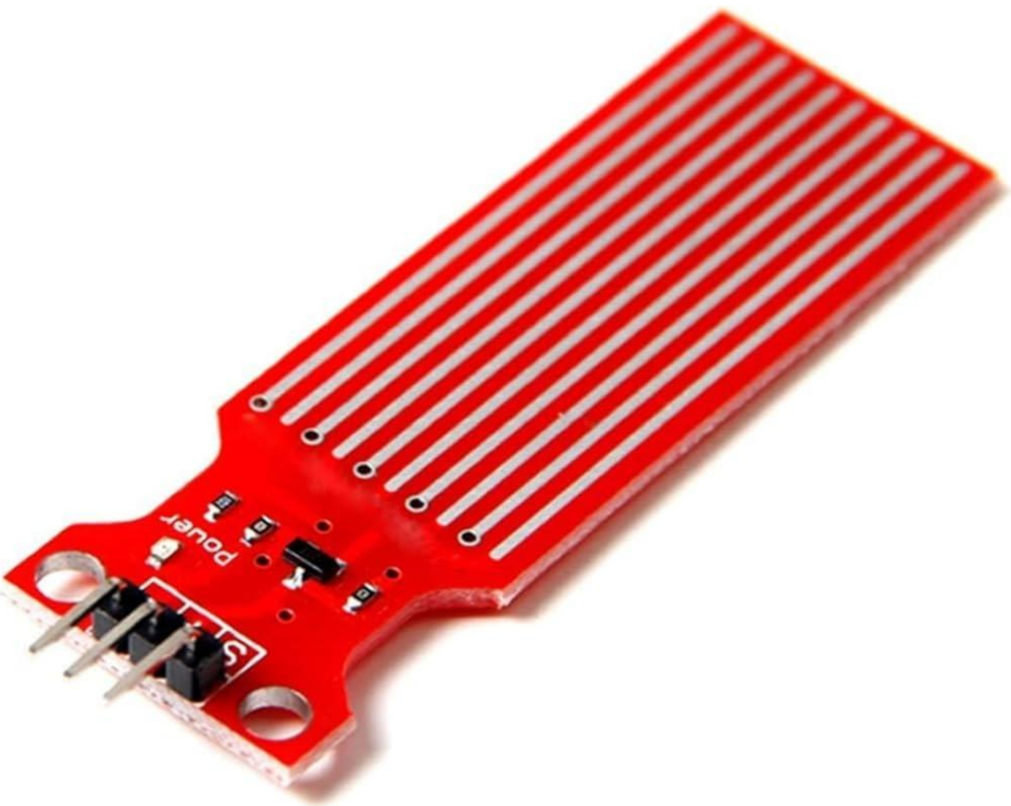

Fig. Water Sensor

- **Accelerometer Sensor:**

An accelerometer is a sensor used to measure acceleration, tilt, and movement of an object in different directions. It can detect changes in position, orientation, and sudden movements such as falls. In a smart walking cane, the accelerometer plays an important role in fall detection by sensing abnormal motion or sudden impact, which can then trigger emergency alerts to caregivers, ensuring quick assistance and enhanced user safety.

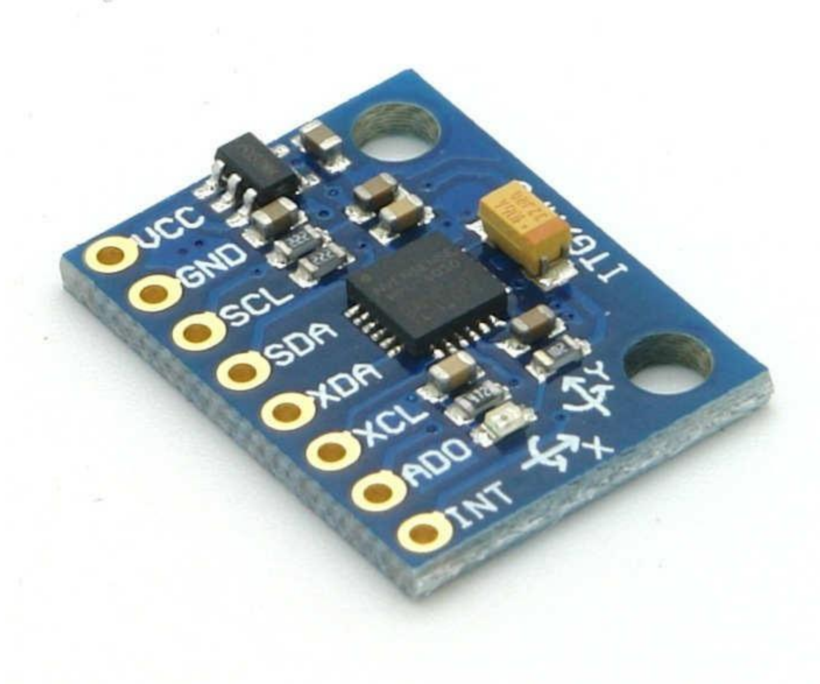

Fig. Accelerometer Sensor

3.4.2 PROCESSING UNIT (ESP32 MICROCONTROLLER)

The ESP32 microcontroller acts as the brain of the system. It performs several important functions:

- Reads data from all sensors
- Processes the data based on predefined thresholds
- Identifies whether a hazard is present
- Controls the buzzer for alerts
- Sends data to the cloud using Wi-Fi

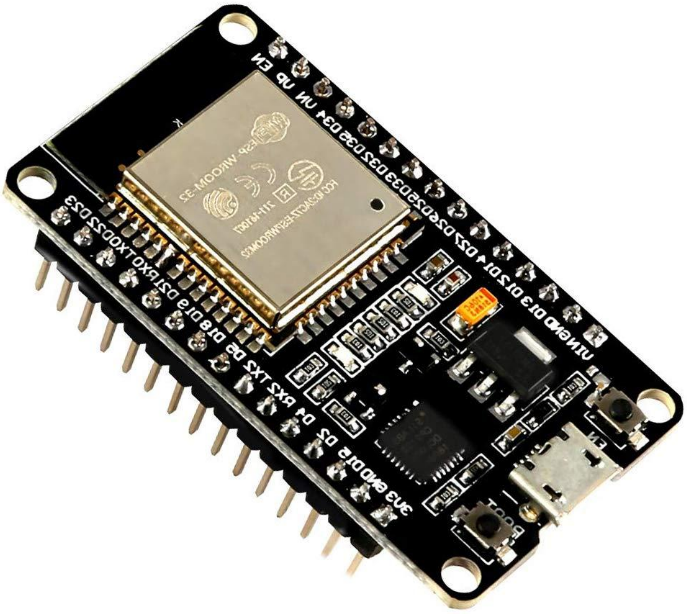

Fig. ESP32

3.4.3 ALERT UNIT

The alert unit provides feedback to the user in real-time.

- A **buzzer** is used to generate sound alerts
- When a hazard is detected, the buzzer is activated
- The alert helps the user take immediate action

This ensures quick response and improves safety.

Fig. Vibration sensor

Fig. Buzzer

3.4.4 GPS MODULE

The GPS module is used for location tracking.

- Provides real-time latitude and longitude
- Helps in tracking the user’s position
- Useful for monitoring and emergency situations

The GPS data is also sent to the cloud for remote access.

Fig. GPS Module

3.4.5 COMMUNICATION UNIT (CLOUD INTEGRATION)

The communication unit allows the system to send data to the cloud using Wi-Fi.

- The ESP32 connects to the internet
- Sensor data and GPS location are sent to cloud platforms such as Firebase
- Enables real-time monitoring

This feature allows caregivers to track the user and monitor system status.

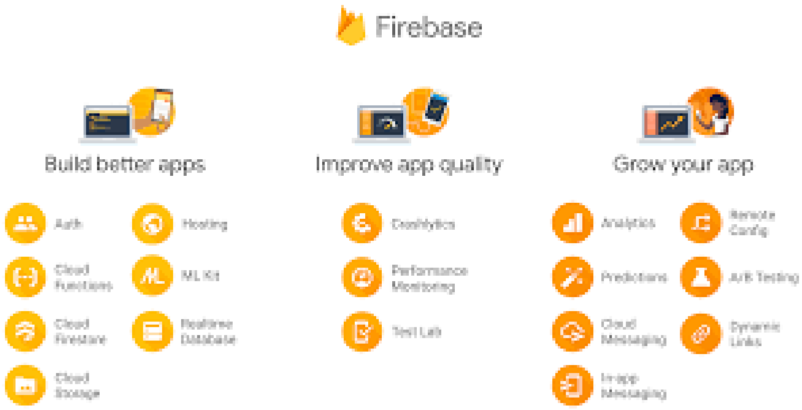

Fig. Firebase

3.4.6 MOBILE APPLICATION DEVELOPMENT

A mobile application was developed using MIT App Inventor to monitor the real-time location of the user. The application retrieves GPS data stored in Firebase and displays it on the interface. This enables caregivers to track the user’s location remotely, improving safety and usability of the system.

- App developed using MIT App Inventor
- Connected with Firebase
- Used for: Location tracking, Viewing GPS data and Monitoring system.

Fig. MIT app inventer

3.5 DATA FLOW IN THE SYSTEM

The data flow in the system can be explained as follows:

1. Sensors collect data from the environment
2. Data is sent to the ESP32 microcontroller
3. ESP32 processes the data
4. If hazard detected → Alert triggered
5. GPS updates location
6. Data sent to cloud
7. System continues monitoring

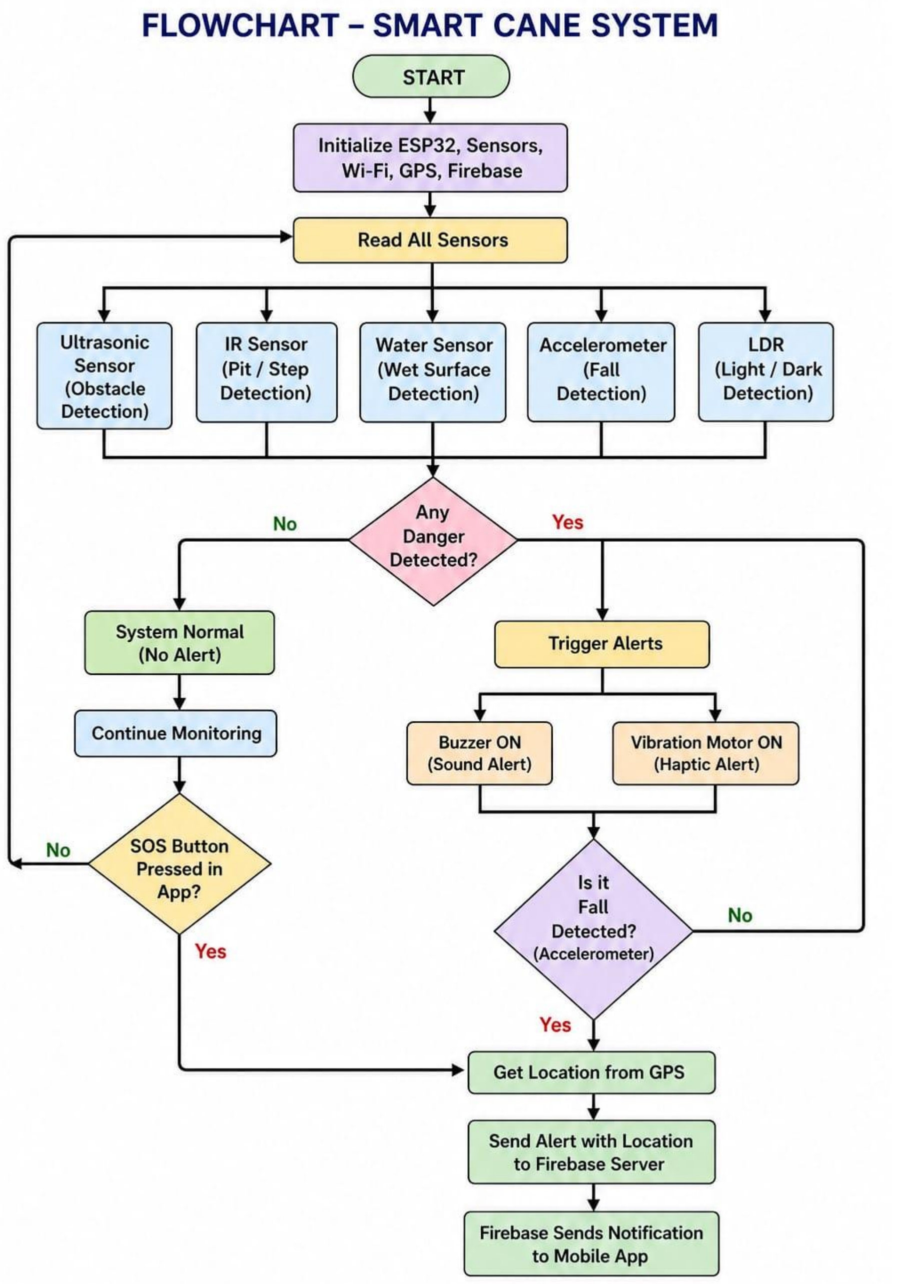

Fig. Flow chart

3.6 WORKING PRINCIPLE

The working of the system is continuous and automatic:

- When the system is powered ON, all sensors and modules are initialized
- Sensors continuously monitor surroundings
- The ESP32 checks sensor values against predefined thresholds
- If any unsafe condition is detected: The buzzer is activated
- GPS continuously updates location
- Data is transmitted to the cloud
- The cycle repeats in real-time

CHAPTER 4

RESULTS

4.1 INTRODUCTION

This chapter presents the experimental results obtained from the implementation of the IoT-based smart walking cane system. The system was tested under different conditions to evaluate its performance in obstacle detection, hazard identification, GPS tracking, and mobile application integration. The results demonstrate the effectiveness and reliability of the proposed system.

4.2 HARDWARE IMPLEMENTATION RESULT

Fig. Model

The above figure shows the developed smart cane prototype. The system is built using a PVC pipe structure with sensors mounted at appropriate positions. The ultrasonic sensor is placed at the front to detect obstacles, while the electronic components are enclosed near the handle.

Observation

- The system is lightweight and portable
- Sensors are properly positioned for effective detection
- Hardware assembly is stable and functional

5.3 MOBILE APPLICATION OUTPUT (LOCATION TRACKING)

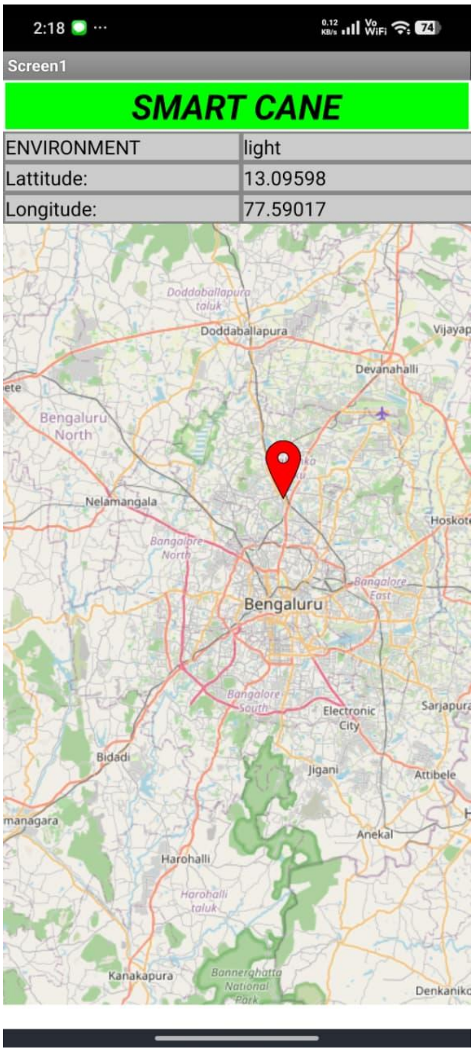

Fig. Location tracking

The mobile application developed using MIT App Inventor successfully displays real-time location data retrieved from Firebase. The application shows latitude, longitude, and a map with a marker indicating the user's position.

Observation

- GPS coordinates are updated correctly
- Map visualization helps in easy tracking
- Data synchronization between hardware and app is successful

5.4 FALL DETECTION ALERT RESULT

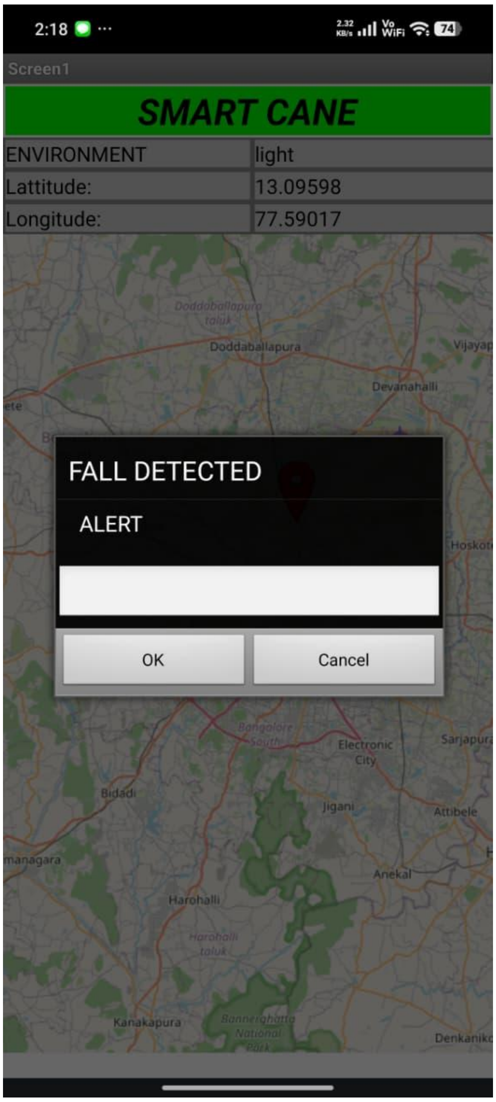

Fig. Alert

The system successfully detects fall conditions and triggers an alert in the mobile application. The alert message "Fall Detected" is displayed, notifying the user or caregiver about the emergency.

Observation

- Alert is generated instantly
- Clear notification improves safety
- Useful for emergency response

4.5 SENSOR PERFORMANCE

Obstacle Detection

- Ultrasonic sensor detects obstacles within predefined range
- Accurate detection observed for short distances

Hazard Detection

- IR sensor detects pits or edges
- Water sensor detects wet surfaces effectively
- LDR identifies low-light conditions

4.6 CLOUD COMMUNICATION RESULT

- Data is successfully transmitted to Firebase
- Real-time updates observed
- Reliable communication between ESP32 and cloud

5.7 DISCUSSION

The results show that the system performs efficiently in detecting obstacles and hazards. The integration of multiple sensors improves accuracy and reliability. The mobile application enhances usability by providing real-time location tracking and alerts.

The system demonstrates:

- Fast response time
- Reliable sensor performance
- Effective cloud communication

Limitations

- GPS accuracy is reduced indoors
- Minor delay in cloud updates
- Sensor calibration required for better accuracy

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The project titled “IoT-Based Smart Walking Cane for Obstacle Detection and Safety Enhancement” was successfully designed and implemented to assist visually impaired individuals in safe navigation. The system integrates multiple sensors such as ultrasonic, IR, LDR, and water sensors to detect obstacles and environmental hazards in real time. The ESP32 microcontroller processes the sensor data efficiently and provides immediate feedback through a buzzer, enabling users to respond quickly to potential dangers. The inclusion of a GPS module enhances the system by providing real-time location tracking, which can be monitored remotely through cloud platforms such as Firebase. This feature improves safety by allowing caregivers to track the user’s location. The system also supports event logging and cloud data storage, making it useful for monitoring and analysis. The developed system is cost-effective, portable, and energy-efficient. It successfully meets the objectives of early hazard detection, real-time alert generation, and remote monitoring. The integration of IoT technology with assistive devices demonstrates a practical solution for improving the independence and safety of visually impaired individuals. Overall, the project provides a reliable and scalable approach to assistive navigation and can be further enhanced with advanced technologies.

7.2 FUTURE SCOPE

Although the proposed system performs effectively, several improvements can be made to enhance its functionality:

- **Voice Assistance:** Integration of audio guidance to provide spoken alerts instead of only buzzer sounds
- **AI-Based Object Detection:** Use of machine learning or computer vision to identify different types of obstacles
- **Mobile Application Development:** A dedicated mobile app for better visualization and control of data
- **Improved Power Management:** Use of efficient batteries and power optimization techniques
- **Miniaturization:** Reducing size and improving portability of the device
- **Indoor Navigation Support:** Integration of technologies such as Bluetooth beacons for indoor tracking
- **Advanced Alert System:** Using vibration motors along with sound for better feedback

REFERENCES

1. R. R. Khandare and S. Patil, "Smart Walking Stick for Visually Impaired Using Ultrasonic Sensor," *International Journal of Engineering Research & Technology*, vol. 5, no. 3, pp. 45–48, 2016.
2. A. Kumar and B. Sharma, "IoT-Based Smart Cane for Visually Impaired People," *International Journal of Advanced Research in Computer Science*, vol. 9, no. 2, pp. 120–124, 2018.
3. S. Koley and R. Mishra, "Voice Operated Outdoor Navigation System for Visually Impaired Persons," *IEEE Transactions on Instrumentation and Measurement*, vol. 60, no. 5, pp. 1–8, 2011.
4. M. Bousbia-Salah, A. Redjati, M. Fezari, and M. Bettayeb, "An Ultrasonic Navigation System for Blind People," *IEEE International Conference on Signal Processing*, pp. 1–4, 2007.
5. P. Mehta, S. K. Gupta, and A. Singh, "Smart Stick for Blind People," *International Journal of Scientific & Engineering Research*, vol. 7, no. 10, pp. 150–155, 2016.
6. ESP32 Datasheet, Espressif Systems, 2023. [Online]. Available: https://www.espressif.com
7. Ultrasonic Sensor HC-SR04 Datasheet, Elecfreaks, 2022.
8. NEO-6M GPS Module Datasheet, u-blox AG, 2022.
9. "Firebase Realtime Database Documentation," Google, 2024. [Online]. Available: https://firebase.google.com
10. "MIT App Inventor Documentation," Massachusetts Institute of Technology, 2024. [Online]. Available: https://appinventor.mit.edu
11. V. R. Reddy and P. R. Kumar, "Smart Navigation System for Visually Impaired Using IoT," *International Journal of Innovative Technology and Exploring Engineering*, vol. 8, no. 6, pp. 123–127, 2019.
12. S. Sharma and R. Gupta, "Obstacle Detection System for Visually Impaired Using Sensors," *International Journal of Computer Applications*, vol. 162, no. 2, pp. 20–24, 2017.
13. A. N. Ansari and S. Shaikh, "IoT-Based Assistive Device for Blind People," *International Conference on Smart Systems and Inventive Technology*, pp. 1–5, 2018.
14. "Arduino IDE Documentation," Arduino, 2024. [Online]. Available: https://www.arduino.cc
15. B. Singh and M. Kaur, "Design of Smart Cane with Multi-Sensor Integration for Blind People," *International Journal of Engineering Science and Computing*, vol. 7, no. 5, pp. 110–115, 2017.